Graphs, Networks, Incidence Matrices

Turn a network into a matrix and its four subspaces start talking. On a network of Nifty spot and futures prices: spreads are edges, only relative prices matter, loops are the left null space, and a loop of quotes that doesn't add to zero is an arbitrage.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Conditions and important concepts**
   12 short definitions: directed graphs, the incidence matrix, differences, loops, Kirchhoff's law, Euler's formula, arbitrage.
2. **Procedure**
   From a network to its matrix, its loops and the arbitrage test.
3. **2 × 2 example**
   The same spread quoted on two venues.
4. **3 × 3 example**
   A triangle: spot, October and November futures.
5. **Bigger example: four contracts, five spreads**
   Two independent loops, a mispriced quote and the arbitrage trade. With a spread checker.
6. **Connection: the four subspaces of a graph**
   Dimensions $n-1$, 1, $n-1$, $m-n+1$, and Lesson 10's test.
7. **Final memory table**
   Everything on one screen.

METHODS IN THIS LESSON: Incidence matrix · Grounding a node · Trees and loops · Kirchhoff's current law · Euler's formula · Loop arbitrage test

PART 2 Prerequisites

Click any link to review it, then come back.

- The four subspaces and the left null space test (Lesson 10).
- Graphs (Lesson 11).

PART 3

Explanation

THE NETWORK USED IN THE EXAMPLES

Nodes (contracts): Nifty spot, October future, November future, December future. Their prices (in points): spot 20,000, Oct 20,100, Nov 20,250, Dec 20,400.

Edges (quoted spreads): a spread from contract a to contract b is "price of b minus price of a ". Traders quote spreads directly, and a set of quotes may or may not be consistent.

1. Conditions and important concepts

1. A directed graph

A graph is nodes joined by edges (Lesson 11). Put an **arrow** on each edge to give it a direction. Here an arrow from spot to Oct means "the Oct − spot spread".

2. The incidence matrix

DEFINITION · INCIDENCE MATRIX A

One **row per edge**, one **column per node**. If an edge runs from node a to node b, its row has -1 in column a, $+1$ in column b, and 0 elsewhere. With m edges and n nodes, A is $m \times n$.

3. Ax = differences across the edges

Let x list the prices at the nodes. Each row of A picks out (end price) − (start price), so

$$\boxed{Ax = \text{the spread on every edge.}}$$

4. The null space: only relative prices matter

$Ax = 0$ means every spread is zero, so all connected prices are equal: $x = c(1, 1, \ldots, 1)$. For a connected graph, $\dim N(A) = 1$. In money terms: **add the same amount to every price and no spread changes.** Spreads alone can never tell you the price *level*.

5. Grounding a node

> METHOD · GROUND ONE NODE
>
> Fix one price (e.g. spot $= 20{,}000$) as the reference. Then the spreads decide every other price. This is the "ground" in an electrical circuit, and the **numeraire** in finance.

So the rank is $r = n - 1$: n prices minus the one free level.

6. Loops make rows dependent

Going around a loop, the spreads must add up: Oct − spot plus Nov − Oct equals Nov − spot. So the row for the edge spot→Nov is the **sum** of the rows for spot→Oct and Oct→Nov. Loops in the graph = dependent rows of A.

7. Trees

> RULE · A TREE = INDEPENDENT ROWS
>
> A **tree** is a set of edges that connects every node with no loop. It has exactly $n - 1$ edges, its rows of A are independent, and its spreads alone decide every price (once one node is grounded).

8. The left null space: positions that cancel

Hold y_i lots of spread i (positive = buy the spread: long its end contract, short its start contract). Then $A^T y$ = your **net position in each contract**. $A^T y = 0$ means you are flat in every contract: the spread positions cancel completely.

> RULE · KIRCHHOFF'S CURRENT LAW: A^T Y = 0
>
> In a circuit, y is the currents and $A^T y = 0$ says current in = current out at every node. Here it says lots in = lots out at every contract. Its solutions are flows **around loops**, and the independent loops are a basis of $N(A^T)$.

9. Counting loops: Euler's formula

$\dim N(A^T) = m - r = m - (n - 1)$, and that is the number of independent loops.
Rearranged:

RULE · EULER'S FORMULA (CONNECTED GRAPH)

$$\text{nodes} - \text{edges} + \text{loops} = 1$$

10. When are quoted spreads consistent?

Quotes e come from real prices exactly when $e = Ax$ for some x, i.e. $e \in C(A)$. By Lesson 10's test, that happens exactly when $y^T e = 0$ for every y in $N(A^T)$:

$$\boxed{\text{Quotes are consistent} \iff \text{they add to zero around every loop.}}$$

11. Arbitrage

RULE · THE LOOP ARBITRAGE TEST

If the quotes around some loop add to something non-zero, the loop vector y is a trade that is flat in every contract ($A^T y = 0$) yet locks in $y^T e$ points. That is an **arbitrage**. No-arbitrage $\iff$ quotes in $C(A)$ $\iff$ every loop sums to zero.

12. The four subspaces at a glance

Subspace	Dimension	Meaning here
$C(A)$	$n - 1$	spread quotes that are consistent (no arbitrage)
$N(A)$	1	shift every price by the same amount
$C(A^T)$	$n - 1$	net positions you can build from spreads (they always sum to zero)
$N(A^T)$	$m - n + 1$	loops: spread positions that cancel out

2. Procedure

Step	What to do
1	Number the nodes and edges; give each edge a direction.
2	Write A: for edge $a \to b$, put -1 in column a and $+1$ in column b.
3	Null space: all-equal vectors $c(1, \dots, 1)$. Rank $r = n - 1$.
4	Loops: find $m - n + 1$ independent loops; write each as y (+1 along the arrow, −1 against). Check $A^T y = 0$.
5	Test the quotes: add them around each loop ($y^T e$).
6	All zero → consistent: ground one node and read off every price. Any non-zero → arbitrage: trade the loop.

3. 2 × 2 example

Two nodes (spot, Oct) and two edges: the Oct − spot spread quoted on two venues, NSE (edge 1) and a second venue (edge 2). Both run spot → Oct:

$$A = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \begin{matrix} \leftarrow \text{edge 1: NSE quote} \\ \leftarrow \text{edge 2: venue 2 quote} \end{matrix}$$

- **Null space:** row 2 − row 1 $= (0, 0)$, so rank $1 = n - 1$. $N(A)$: $(1, 1)$, shift both prices.
- **Loop:** $m - n + 1 = 2 - 2 + 1 = 1$. Buy the spread on venue 1, sell it on venue 2: $y = (1, -1)$. Check $A^T y = (-1 \cdot 1 + (-1)(-1),\ 1 \cdot 1 + 1 \cdot (-1)) = (0, 0)$ ✓: flat in spot and Oct.
- **Consistent quotes:** $(100, 100)$: $y^T e = 100 - 100 = 0$ ✓.
- **Inconsistent quotes:** $(100, 105)$: $y^T e = 100 - 105 = -5 \neq 0$. Arbitrage: buy the spread where it's 100, sell it where it's 105, lock in **5 points**.

Euler: $2 - 2 + 1 = 1$ ✓.

4. 3 × 3 example

Three nodes (spot, Oct, Nov) and three edges: spot→Oct, Oct→Nov, spot→Nov.

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{matrix} \leftarrow \text{spot} \rightarrow \text{Oct} \\ \leftarrow \text{Oct} \rightarrow \text{Nov} \\ \leftarrow \text{spot} \rightarrow \text{Nov} \end{matrix} \qquad Ax = \begin{bmatrix} x_{\text{Oct}} - x_{\text{spot}} \\ x_{\text{Nov}} - x_{\text{Oct}} \\ x_{\text{Nov}} - x_{\text{spot}} \end{bmatrix}$$

- **Dependent rows:** row 3 = row 1 + row 2: $(-1+0,\ 1-1,\ 0+1) = (-1, 0, 1)$ ✓. That's the loop.
- **Rank** $= 2 = n - 1$. **Null space:** $(1, 1, 1)$.
- **Loop** $(3 - 3 + 1 = 1)$: $y = (1, 1, -1)$, buy spot→Oct and Oct→Nov, sell spot→Nov. $A^T y$: spot $-1 + 0 + 1 = 0$, Oct $1 - 1 + 0 = 0$, Nov $0 + 1 - 1 = 0$ ✓.
- **Consistent quotes** from the prices: $(100, 150, 250)$. Loop: $100 + 150 - 250 = 0$ ✓.
- **A mispriced quote:** spot→Nov quoted at 240. Loop: $100 + 150 - 240 = 10 \neq 0$. The two short spreads say Nov − spot $= 250$, the direct quote says 240: buy the direct spread at 240, sell the two-step route at 250, flat in every contract, **10 points** locked in.

5. Bigger example: four contracts, five spreads

Nodes 1–4: spot, Oct, Nov, Dec. Edges:

Edge	From → to	Row of A	Spread from prices
1	spot → Oct	$(-1, 1, 0, 0)$	$20{,}100 - 20{,}000 = 100$
2	Oct → Nov	$(0, -1, 1, 0)$	$20{,}250 - 20{,}100 = 150$
3	spot → Nov	$(-1, 0, 1, 0)$	$20{,}250 - 20{,}000 = 250$
4	spot → Dec	$(-1, 0, 0, 1)$	$20{,}400 - 20{,}000 = 400$
5	Nov → Dec	$(0, 0, -1, 1)$	$20{,}400 - 20{,}250 = 150$

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \qquad m = 5 \text{ edges},\ n = 4 \text{ nodes}$$

- **Null space:** $(1, 1, 1, 1)$. Rank $r = n - 1 = 3$.
- **A tree:** edges 1, 2, 4 (spot→Oct, Oct→Nov, spot→Dec) connect all four contracts with no loop. With spot grounded at 20,000, they give Oct $= 20{,}100$, Nov $= 20{,}250$, Dec $= 20{,}400$.
- **Loops:** $m - r = 5 - 3 = 2$. Euler: $4 - 5 + 2 = 1$ ✓.

Loop	y (lots of each spread)	Check $A^T y$ (spot, Oct, Nov, Dec)	Quotes around it
spot→Oct→Nov, back via spot→Nov	$(1, 1, -1, 0, 0)$	$(-1+1,\ 1-1,\ 1-1,\ 0) = (0,0,0,0)$ ✓	$100 + 150 - 250 = 0$ ✓
spot→Nov→Dec, back via spot→Dec	$(0, 0, 1, -1, 1)$	$(-1+1,\ 0,\ 1-1,\ -1+1) = (0,0,0,0)$ ✓	$250 + 150 - 400 = 0$ ✓

The outer loop (spot→Oct→Nov→Dec, back via spot→Dec) is $(1, 1, 0, -1, 1)$, the **sum** of the two: not a new independent loop.

A mispriced quote

The market quotes spot→Nov at **240** instead of 250. Test both loops:

$$\text{loop 1: } 100 + 150 - 240 = \boxed{10} \qquad \text{loop 2: } 240 + 150 - 400 = \boxed{-10}$$

Both flag the same bad quote (edge 3 is in both loops). Loop 1 says: buy the cheap direct spread at 240, sell spot→Oct and Oct→Nov (worth 250 together), and you are flat in every contract with 10 points locked in.

Try it: the spread checker

Edge 1 (Spot→Oct) 100 Edge 2 (Oct→Nov) 150 Edge 3 (Spot→Nov) 240

Edge 4 (Spot→Dec) 400 Edge 5 (Nov→Dec) 150

Incidence matrix (rows = edges, columns = Spot, Oct, Nov, Dec): $A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. Rank $3 = n - 1$; loops: $m - r = 5 - 3 = 2$.

Loop 1: $y = (-1, -1, 1, 0, 0)$. Quotes around it: $-100 - 150 + 240 = -10$ ✗

Loop 2: $y = (1, 1, 0, -1, 1)$. Quotes around it: $100 + 150 - 400 + 150 = 0$ ✓

Arbitrage: loop 1 adds to -10. Trade it: sell 1 × edge 1, sell 1 × edge 2, buy 1 × edge 3. Flat in every contract ($A^T y = 0$), locking in 10 points.

6. Connection: the four subspaces of a graph

Subspace	Dimension	Example (5 edges, 4 nodes)	Meaning
$N(A)$	1	$(1,1,1,1)$	only relative prices matter
$C(A^T)$	$n-1=3$	rows of a tree	net positions buildable from spreads
$C(A)$	$n-1=3$	quotes from real prices	consistent, arbitrage-free spreads
$N(A^T)$	$m-n+1=2$	the two loops	self-cancelling positions; the arbitrage test

The pairs from Lesson 10 hold: $3+1=4=n$ and $3+2=5=m$ ✓.

This is Lesson 10's big idea in its most useful form: **the left null space is the test for which right-hand sides are reachable.** For a network of prices, "reachable" means "could come from real prices", and the test is "every loop adds to zero". Currency triangles work the same way using logarithms (Example 1 on the Examples tab).

▼ For the curious: the electrical version, A^TCA

In a circuit, x is the voltages, $e=Ax$ the voltage drops, $y=Ce$ the currents (Ohm's law, with C the conductances), and $A^Ty=f$ the current balance with sources f. Together: $A^TCAx=f$. MIT's square graph with $C=\mathrm{diag}(1,2,2,1)$ gives $A^TCA=\begin{bmatrix}2&-1&0&-1\\-1&3&-2&0\\0&-2&4&-2\\-1&0&-2&3\end{bmatrix}$: symmetric, as A^TCA always is.

7. Final memory table

Concept	Key fact	In the spread network
Incidence matrix	row per edge: -1 at start, $+1$ at end	Ax = spreads from prices
Null space	$c(1,\dots,1)$; dim 1	shifting all prices changes no spread
Rank	$n-1$	ground one node (numeraire)
Tree	$n-1$ edges, no loop, independent rows	enough spreads to price everything
Loops	basis of $N(A^T)$; $m-n+1$ of them	spread positions that cancel out
Kirchhoff	$A^Ty=0$	flat in every contract
Euler	nodes $-$ edges $+$ loops $=1$	$4-5+2=1$
No arbitrage	$e\in C(A) \iff y^Te=0$ for every loop	quotes add to zero around every loop

▼ **Check yourself (MIT problem 12.1): the square graph with edges 1→2, 2→3, 4→3, 1→4. What is $N(A)$? Why is $(1,0,0,0)$ not in the row space?**

$A=\begin{bmatrix}-1&1&0&0\\0&-1&1&0\\0&0&1&-1\\-1&0&0&1\end{bmatrix}$. $Ax=0$ forces $x_1=x_2=x_3=x_4$, so $N(A)$ = all (a,a,a,a). Every row of A adds to zero, so every vector in the row space adds to zero; $(1,0,0,0)$ adds to 1, so it isn't there. (In money terms: spreads can only build positions that are net flat.)